

TriSaFe-Trans: A Safety-Aware Multimodal Intent Recognition Pipeline for Assistive Robotics

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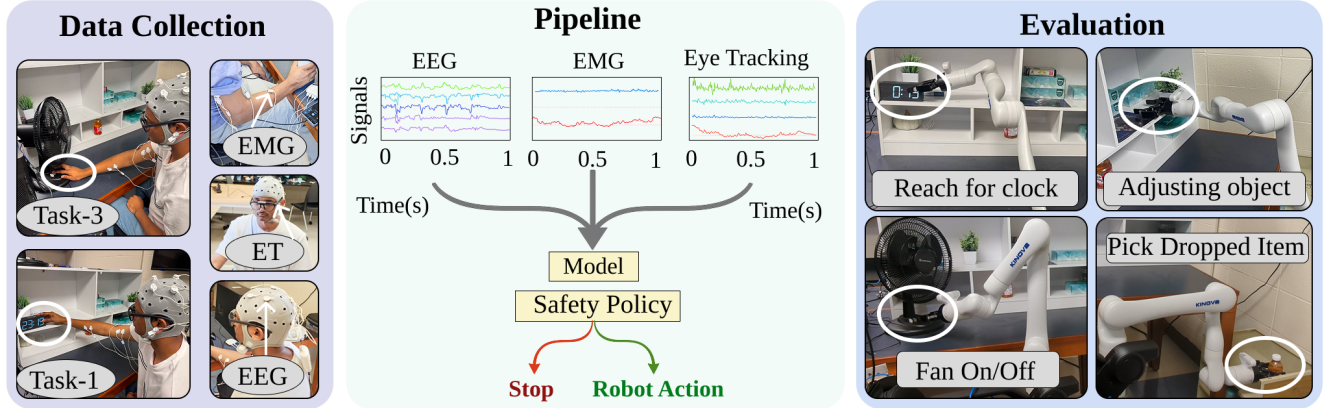


Fig. 1. TriSaFe-Trans pipeline. **Left:** Synchronized EEG, EMG, and eye-tracking data collected during assistive tasks, capturing complementary neural, muscular, and gaze-based intent cues. **Middle:** A tri-modal model decodes action and task intent from aligned sensor windows and applies a safety policy that gates robot execution by selecting between action and stop. **Right:** Demonstration on representative assistive manipulation tasks, showing translation of inferred intent into physical robot actions.

Abstract— Safe intent recognition for assistive robots remains challenging when integrating neural, muscular, and gaze signals, particularly under sensor degradation or failure. We propose TriSaFe-Trans, a lightweight tri-modal transformer that fuses electroencephalography (EEG), electromyography (EMG), and eye-tracking (ET) data via modality-specific temporal convolutional networks (TCNs) and bidirectional gated recurrent units (BiGRUs), combined with reliability-gated cross-modal attention. The model is trained using multi-objective self-supervised pretraining, followed by supervised fine-tuning, to simultaneously decode action (REST vs. ACTION) and task (rest plus five assistive tasks) from fixed-length windows. We collected a synchronized EEG, EMG, and eye-tracking dataset from $N=20$ healthy adults, and evaluated the model using a balanced-window leave-one-subject-out (LOSO) protocol. Under nominal sensing (S0), TriSaFe-Trans achieves an action accuracy of 0.903 ± 0.079 (balanced accuracy 0.882 ± 0.079) and a task accuracy of 0.727 ± 0.096 (balanced accuracy 0.680 ± 0.100), with positive multimodal synergy gain. Across LOSO folds, TriSaFe-Trans improves task decoding and shows statistically significant gains over the baselines on key metrics. We assess single-modality failures via test-time sensor-dropout scenarios S1 (drop EEG), S2 (drop EMG), and S3 (drop ET) masking the dropped modality, and introduce safety policies (P0-P2) with safety metrics to quantify safety performance trade-offs. Under S0, a safety-first policy reduces unsafe actions and improves rest safety at the cost of increased missed intent. Code and supplementary video are available at: <https://github.com/madibabaiasl/SafeIntentDetectionPaper/wiki> and https://youtu.be/pdMp_myF-1I, respectively.

I. INTRODUCTION

Wheelchair-mounted robotic arms (WMRAs) are an emerging assistive technology with substantial potential to enhance user independence by enabling activities of daily living (ADLs) that may be difficult or impossible for the user without assistance, including basic tasks such as reaching and grasping [1]. Because these systems physically interact with humans, safety and dependability are the core design requirements in physical human-robot interaction (pHRI) systems [2], [3]. This motivates the development of intent interfaces that reduce dependence on conventional controls by leveraging non-invasive EEG, surface EMG, and eye-tracking. In assistive robotics, conventional interfaces such as joysticks and body-machine control mappings remain important practical baselines. Rather than claiming to replace them universally, our goal is to study when multimodal physiological sensing can complement them and reduce user burden during assistive manipulation. EEG-based brain-computer interfaces (BCIs) decode motor-related cortical activity [4], EMG captures residual muscle activation for myoelectric control [5], and gaze can suggest the intended goal of the user in shared autonomy or manipulation scenarios [6].

However, many intent decoders are evaluated only as offline window-level classification (accuracy/F1) [7], [8], even though deployment requires an execute/stop decision where false positives during REST can trigger unintended activations and conservative thresholds can miss true intents. Moreover, EMG signals are vulnerable to nonideal sensing and artifacts [9] and eye-tracking streams inherently contain data gaps due to blinks, while fixed/explicit fusion can be

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brittle when a modality becomes unreliable or missing [10], [11]; thus, a key gap is reliability-aware fusion with policy-level evaluation that quantifies unsafe actions versus missed intent under sensor failures.

Existing public datasets for upper-limb intent decoding typically focus on EEG-EMG signals for isolated gestures or simple movements (e.g., hand/forearm gestures classes [12], wrist/fist actions [13]) or standardized reaching paradigms [14], while other EEG datasets target unrelated paradigms such as auditory-evoked biometrics [15]. In contrast, the dataset introduced in this work is tailored to assistive manipulation: it provides *synchronized tri-modal* EEG-EMG-eye-tracking recordings during ADL-style tasks (T1-T5) with explicit REST periods, enabling subject-independent LOSO evaluation and controlled sensor-dropout tests (S0-S3). These gaps motivate the following question: *Can a safety-aware tri-modal transformer that fuses EEG, EMG, and eye-tracking, evaluated at the policy level and tested under sensor failures, reduce unsafe actions and missed assistance compared to unimodal and naive multimodal baselines?* This study addresses this question with TriSaFe-Trans (see **Figure 1**). The main contributions are as follows:

- A safety-aware tri-modal transformer for EEG, EMG, and eye-tracking, using lightweight per-modality TCN+BiGRU encoders, reliability-gated cross-modal fusion, and hybrid deep and engineered EEG/EMG features for joint action and task decoding.
- Policy-level safety and robustness evaluation using the unsafe fraction among executed actions (SAE), the correct execution rate on ACTION windows (UAR), missed intention rate (MIR), and rest-suppression (safe-idle) rate (NRA) across policies P0–P2 and sensor-failure scenarios S0–S3, complemented by Multimodal Synergy Gain (MSG) and Global Modality Usage Entropy (GMUE).
- Release of, to our knowledge, the first synchronized EEG-EMG-eye-tracking dataset for functionally assistive ADL (International Classification of Functioning, Disability, and Health (ICF)-aligned) intent decoding, benchmarked with a balanced-window LOSO protocol.

II. RELATED WORK

Non-invasive BCIs use EEG to decode motor imagery or event-related activity into commands for communication and device control [16]. Invasive BCIs have demonstrated direct robotic arm control in people with tetraplegia [17]. In parallel, surface EMG is widely used to control upper-limb prostheses, and multi-channel EMG pattern recognition is a prominent approach for discriminating multiple grasps or joint motions by mapping EMG patterns to motion classes [18]. However, many EEG/EMG intent studies still exclude policy-level safety outcomes (e.g., unintended rest activations and missed assistance) in their evaluation.

Eye gaze is a hands-free signal for assistive teleoperation and shared autonomy, as it often indicates the object or region a user intends to act upon. Prior studies have utilized gaze

features to estimate intention in shared autonomy manipulation, reporting improved efficiency and reduced operator load [6], [19]. However, many gaze-based controllers rely on simple heuristics (e.g., dwell time or thresholds) and rarely account for failures, calibration drift, or uncertainty when deciding whether to take action.

Fusing EEG and EMG signals can improve stability and performance relative to single-modality control [20], [21], and can be more resilient under fatigue [20]. However, fusion robustness depends on the fusion strategy; static fusion can fail when the signal quality of one modality changes substantially, motivating reliability-aware (dynamic) weighting of modalities [11]. Hybrid assistive interfaces often combine EEG with another physiological channel (e.g., EMG or EOG) or with an external sensor such as eye-tracking [22], [23]. At the same time, tri-modal EEG-EMG-ET pipelines remain comparatively rare. Separately, pHRI safety research has advanced compliant actuation, collision handling, and injury-aware control [2], and standards-oriented studies emphasize safeguards grounded in hazard analysis and risk assessment [24]. However, these frameworks typically do not incorporate tri-modal intent decoders, and intent-recognition studies rarely evaluate safety at the policy level. This study connects these strands by introducing a tri-modal EEG-EMG-eye-tracking model with reliability-gated fusion and by analyzing explicit safety-performance trade-offs under sensor-failure scenarios using risk-oriented metrics.

III. METHODS

An end-to-end pipeline was used to convert synchronized EEG, EMG, and ET streams into fixed-length labeled windows and policy-level execute/stop decisions (**Figure 2**). After alignment and deterministic preprocessing, action (REST vs. ACTION) and task labels were assigned to each window. TriSaFe-Trans was then trained, and three safety policies were evaluated under both nominal sensing and simulated modality dropouts: P0 uses the raw action probability, P1 scales it by sensor reliability, and P2 further down-weights it by prediction uncertainty, making P2 the most conservative.

A. Dataset and Experimental Protocol

A synchronized tri-modal dataset was collected from 20 adults (12 male/8 female; age 25.50 ± 6.79 years (mean $\pm$ SD)) in a standard laboratory environment using a chair, table, and shelf setup. Each session lasted approximately 90–120 minutes, including sensor setup, calibration, task execution, and breaks, totaling around 35–40 participant-hours. The study was approved by the Institutional Review Board (IRB) of Saint Louis University (Assurance No. FWA00005304), and all participants provided written informed consent.

Six intent classes are considered: REST and five ADL-style tasks (T1-T5). T1 (Clock) requires reaching toward a tabletop clock and grasping it. T2 (Bottle) involves picking up a dropped apple juice bottle and placing it back onto the table. T3 (Fan) consists of pressing the fan power button to toggle it on and off. T4 (Plant) requires adjusting or

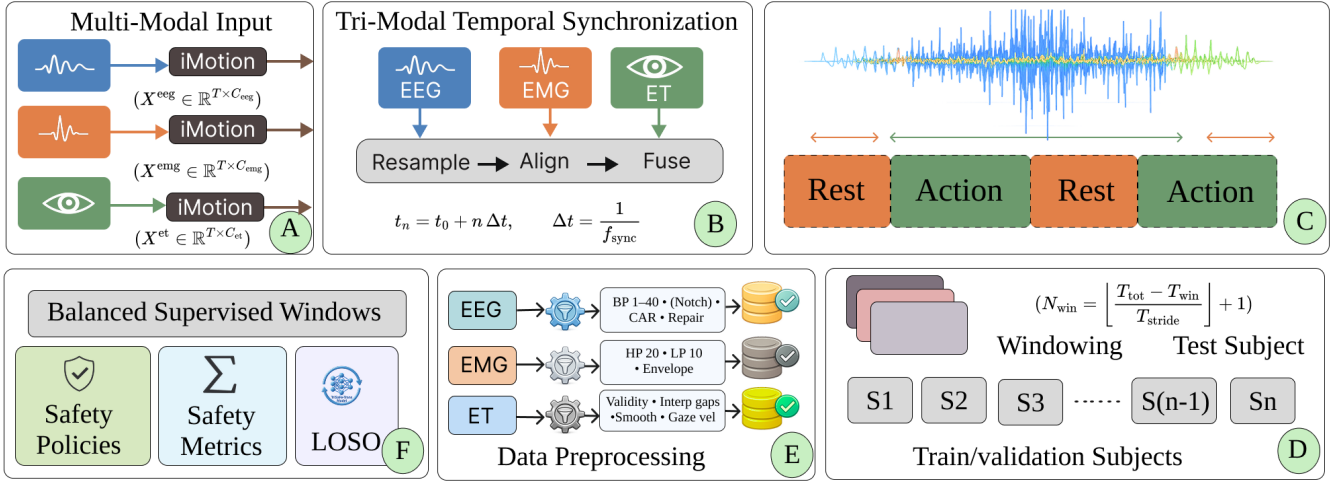


Fig. 2. Tri-modal safety-aware data processing and evaluation pipeline. Raw EEG, EMG, and eye-tracking streams are acquired and temporally synchronized (A–B) to enable aligned multimodal analysis. A hidden semi-Markov model (HSM) segments continuous data into REST and ACTION states (C), providing robust intent boundaries. The synchronized signals are windowed and split using a leave-one-subject-out (LOSO) protocol (D) to support subject-independent evaluation. Deterministic, modality-specific preprocessing ensures consistent and reproducible inputs (E). Balanced supervised windows are then used for safety-aware LOSO training and evaluation with explicit safety policies and metrics (F), enabling policy-level assessment of unsafe actions and missed intent. BP/HP/LP = Band-pass/High-pass/Low-pass filter, CAR = Common Average Referencing, Interp = interpolation, and vel = velocity.

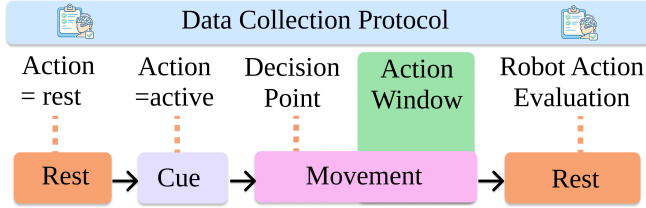


Fig. 3. Experimental protocol and labeling timeline. Each trial consists of pre-trial rest, cue onset, movement execution, and post-movement rest. Action (REST vs. ACTION) labels and safety decision points are defined relative to this timeline, with the action window used for intent decoding and policy evaluation.

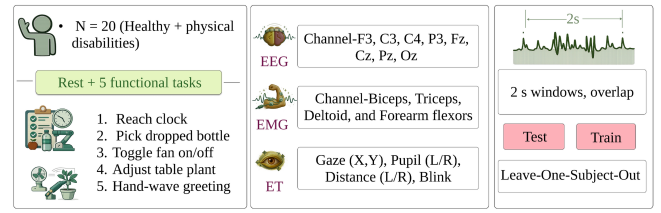


Fig. 4. Tri-modal dataset summary. Overview of participants, assistive tasks, sensing modalities (EEG, EMG, and eye tracking), windowing parameters, and the LOSO protocol used for subject-independent action and task evaluation.

repositioning a plant on the shelf. T5 (Wave) entails performing a greeting gesture by waving the hand. Each participant completed 16 trials per active task. Each trial followed a fixed structure: (i) pre-trial rest, (ii) visual or verbal cue, (iii) movement execution, and (iv) post-movement rest (**Figure 3**). The protocol provided both REST-labeled and ACTION-labeled segments. After HSM-based segmentation, model windows were extracted from labeled segments, with ACTION windows corresponding to task-specific active periods and REST windows representing rest-labeled periods used for execute/stop evaluation.

Figure 4 summarizes the participants, task set, sensing streams, windowing, and LOSO protocol. EEG, EMG, and eye-tracking were streamed concurrently into iMotions via Lab Streaming Layer (LSL). EEG was recorded using an Enobio-8 system at 250 Hz, EMG was acquired using a 4-channel wireless PLUX system from upper-limb muscles following SENIAM guidelines [25], [26], and eye-tracking was captured using Pupil Labs Neon glasses. Trial-level task codes from the experiment script were logged in iMotions and synchronized with all sensor streams. Subsequently,

HSM-based consensus pipeline derived fine-grained ACTION vs. REST labels (see **Section III-C**).

B. Robot Execution via Robot Operating System ROS 2

A ROS 2 execution pipeline integrates tri-modal intent inference with a Kinova Gen3 robotic arm and Robotiq gripper. Arm and gripper movements for each assistive task were collected by teleoperating the arm with an Xbox controller and recording joint positions at key positions (e.g., approach, grasp, and release). These joint position sequences are stored as task-specific templates and replayed autonomously by sending FollowJointTrajectory goals to the arm's joint trajectory controller and GripperCommand messages to the gripper controller. During operation, synchronized EEG, EMG, and eye-tracking signals are processed to infer the user's intended task. The predicted task ID selects the corresponding stored trajectory, enabling the robot to execute the intended action without manual teleoperation. Execution is gated by the safety policy layer (P0-P2). If execution is suppressed, no trajectory command is issued and the arm holds position. Otherwise, the selected trajectory and gripper commands are sent for autonomous task completion.

C. Signal Preprocessing and Windowing

A deterministic pipeline was used to convert synchronized EEG, EMG, and eye-tracking streams into model-ready windows used across all experiments. Signals were aligned using Lab Streaming Layer (LSL) timestamps and resampled to 250 Hz. Windows of $T=500$ samples (2.0 s) were extracted with a 0.25 s stride and labeled by majority vote for action (REST vs. ACTION); ACTION windows were assigned the most frequent task label among the six intent classes. LOSO evaluation held out one subject for test and one for validation, with training performed on the remainder. Training windows were stratified and class-balanced across tasks, while validation/test preserved the natural distribution.

EEG was band-pass filtered (1–40 Hz), followed by automatic 50/60 Hz notch filtering when line-noise contamination was detected, common-average referencing, and z -normalization using training-fold statistics; transient artifacts and flat/noisy segments were repaired or tracked using quality masks. EMG was high-pass filtered (> 20 Hz), rectified, low-pass filtered (5–10 Hz) to form an envelope, and z -normalized. Eye-tracking streams were cleaned using blink and validity indicators, short-gap interpolation, and temporal smoothing, while invalid samples were retained through eye-tracking quality masks. Per window, compact EEG Power Spectral Density (PSD)/bandpower and EMG time-domain envelope features (e.g., Root Mean Square (RMS), Mean Absolute Value (MAV), Wave Length (WL)) were computed and z -normalized to form engineered vectors f^{eeg} and f^{emg} . Subjects were indexed as Sub1–Sub20, while sensing conditions were denoted separately as S0–S3. Here, S0 denotes nominal sensing (EEG+EMG+ET), and S1–S3 denote test-time dropout of EEG, EMG, and eye-tracking, respectively, implemented by masking the dropped modality.

D. TriSaFe-Trans Architecture

The proposed TriSaFe-Trans model was a lightweight tri-modal network that encoded EEG, EMG, and ET windows and fused them with feature-wise gated self-attention (**Figure 5**). The fused representation was used for joint action and task prediction, while the learned modality gates were further used to form reliability-aware safety scores.

a) Modality encoders: For each modality $m \in \{\text{eeg}, \text{emg}, \text{et}\}$, an input window $X^m \in \mathbb{R}^{B \times T \times C_m}$ was mapped to a token sequence $Z^m \in \mathbb{R}^{B \times T \times d}$, where B is the batch size, T is the number of temporal samples, C_m is the number of input channels for modality m , and d is the shared embedding dimension. The EEG branch consisted of a lightweight temporal convolutional encoder with three dilated 1D convolution layers, batch normalization, ReLU activations, a residual connection, dropout, a bidirectional Gated Recurrent Unit (GRU), and a linear projection. The EMG branch followed a similar but smaller design, using two dilated 1D convolution layers, a residual connection, dropout, a bidirectional GRU, and a linear projection. The eye-tracking branch used a compact multilayer perceptron (MLP), followed by a bidirectional GRU and a linear projection.

b) Temporal pooling and feature-wise modality gates: After modality encoding, each modality was represented by a token sequence $Z^m \in \mathbb{R}^{B \times T_m \times d}$, where T_m denotes the number of tokens for modality m . To reduce sequence length before fusion, temporal pooling was applied by fixed strided subsampling, $Z^m \leftarrow Z^m[:, :, s, :]$, where s is the temporal pooling stride. A mean-pooled summary was then computed for each modality as $p^m = \frac{1}{T_m} \sum_{t=1}^{T_m} Z^m_{:,t,:}$. The three modality summaries were concatenated and passed through a gating MLP, whose output was reshaped as $G \in \mathbb{R}^{B \times 3 \times d}$. A softmax was then applied across modalities for each feature dimension, $\alpha_{i,m,k} = \frac{\exp(G_{i,m,k})}{\sum_{m'} \exp(G_{i,m',k})}$, where $\alpha_{i,m,k}$ denotes the gate assigned to modality m for sample i and feature dimension k . For interpretability and later policy construction, a mean gate was also computed for each modality as $\gamma_{i,m} = \frac{1}{d} \sum_{k=1}^d \alpha_{i,m,k}$, so that $\gamma_i \in [0, 1]^3$ and $\sum_m \gamma_{i,m} = 1$; these mean gates were later reused by the safety layer.

c) Gated cross-modal fusion: The temporally pooled EEG, EMG, and ET tokens were concatenated with a learnable CLS token to form $X \in \mathbb{R}^{B \times L \times d}$, where L is the total number of modality tokens plus CLS. After positional encoding, each fusion layer applied gated self-attention with $Q = X$, $K = X \odot A$, and $V = X \odot A$, where $A \in \mathbb{R}^{B \times L \times d}$ repeats the corresponding feature-wise modality gate over each token and assigns an all-ones gate to the CLS token. Each block used multi-head attention, residual connection, layer normalization, and a feed-forward network. After fusion, modality-specific summaries were obtained by mean-pooling the corresponding fused tokens, and the final representation was computed as $z_i^{\text{fused}} = \alpha_{i,\text{eeg},:} \odot \bar{z}_i^{\text{eeg}} + \alpha_{i,\text{emg},:} \odot \bar{z}_i^{\text{emg}} + \alpha_{i,\text{et},:} \odot \bar{z}_i^{\text{et}}$, where $\alpha_{i,m,:} \in \mathbb{R}^d$ denotes the feature-wise gate vector for modality m and sample i , and $\bar{z}_i^m \in \mathbb{R}^d$ denotes the post-fusion mean-pooled representation of modality m .

d) Action and task heads: Two prediction heads operated on the fused representation. The action head predicted a binary label (0 = REST, 1 = ACTION), and the task head predicted the assistive task label. Let $o^{\text{act}} \in \mathbb{R}^2$ and $o^{\text{task}} \in \mathbb{R}^K$ denote the action and task logits, respectively. The predicted probability of ACTION was $p^{\text{act}} = \text{softmax}(o^{\text{act}})_1$, and the task posterior was given by $p^{\text{task}} = \text{softmax}(o^{\text{task}})$.

e) Safety signals and execution policies: For a given sensor-availability pattern, the reliability score for sample i was

$$r_i = \sum_{m \in \{\text{eeg}, \text{emg}, \text{et}\}} s_m \gamma_{i,m},$$

where $s_m \in \{0, 1\}$ indicates whether modality m is available and $\gamma_{i,m}$ is the mean gate assigned to modality m .

Task and gate uncertainty were defined by the normalized entropies

$$\begin{aligned} \bar{H}_i^{\text{task}} &= \frac{-\sum_{c=1}^K p_{i,c}^{\text{task}} \log(p_{i,c}^{\text{task}} + \epsilon)}{\log K}, \\ \bar{H}_i^{\text{gate}} &= \frac{-\sum_m \gamma_{i,m} \log(\gamma_{i,m} + \epsilon)}{\log 3}. \end{aligned}$$

where $p_{i,c}^{\text{task}}$ is the predicted probability of task class c , K is the number of task classes, and ϵ is a small constant for

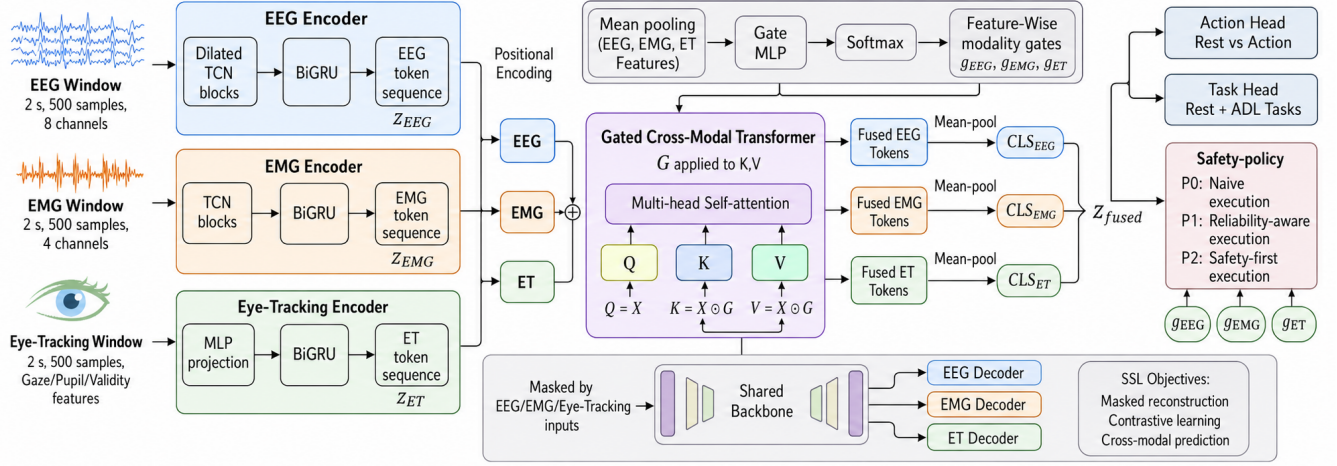


Fig. 5. TriSaFe-Trans architecture. EEG, EMG, and eye-tracking windows were first encoded by modality-specific lightweight encoders. Their token sequences were then fused by a gated cross-modal transformer in which feature-wise modality gates modulated attention. The fused representation was used for joint action and task prediction, and the mean modality gates were reused to construct reliability- and uncertainty-aware safety policies (P0–P2).

numerical stability. The combined uncertainty was

$$u_i = \frac{1}{2} \left(\bar{H}_i^{\text{task}} + \bar{H}_i^{\text{gate}} \right).$$

The three execution policies were defined as follows: P0 used the raw action probability, $p_i^{(0)} = p_i^{\text{act}}$; P1 scaled this probability by reliability, $p_i^{(1)} = p_i^{\text{act}} r_i$; and P2 further penalized uncertainty, $p_i^{(2)} = p_i^{\text{act}} r_i (1 - u_i)$, making it the most conservative policy. Here, p_i^{act} is the predicted probability of ACTION. Execution was triggered when $p_i^{(p)} \geq \tau$; otherwise, the output was set to REST. If execution was triggered, the REST task was suppressed and the highest-scoring non-REST task was selected.

E. Pretraining and Safety-Aware Fine-Tuning

TriSaFe-Trans was trained in two stages. First, self-supervised pretraining was performed on unbalanced windows to learn robust multimodal representations. For each tri-modal sample, two augmented views were generated using random timestep masking and modality dropout. The backbone was optimized with masked reconstruction of EEG, EMG, and eye-tracking tokens, a symmetric contrastive loss between fused embeddings, and a cross-modal prediction loss between modality summaries. EEG and EMG reconstruction were weighted more strongly than eye tracking to emphasize neural and muscular intent cues. The SSL objective was $\mathcal{L}_{\text{ssl}} = \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} + \lambda_{\text{contrast}} \mathcal{L}_{\text{contrast}} + \lambda_{\text{xmod}} \mathcal{L}_{\text{xmod}}$.

Second, the pretrained model was fine-tuned using balanced LOSO training windows with a joint action–task objective. The action head was trained on all windows for binary REST/ACTION classification, while the task head was trained only on ACTION windows to predict the assistive task. The supervised objective was $\mathcal{L}_{\text{sup}} = \alpha_{\text{action}} \mathcal{L}_{\text{action}} + \beta_{\text{task}} \mathcal{L}_{\text{task}}$. During fine-tuning, eye-tracking dropout was applied to improve robustness under partial sensor degradation.

a) *Validation-time threshold tuning and safety evaluation:* The action-decision threshold τ was selected on the validation split by sweeping $\tau \in [0.1, 0.9]$ and maximizing REST-versus-ACTION balanced accuracy, then fixed for test evaluation. Safety was evaluated at the window level using four outcomes: correct move (execution on a true ACTION window with correct task), unsafe move (execution during REST or with an incorrect task), missed intent (no execution during ACTION), and safe idle (suppressed execution during REST). We report $\text{SAE} = \frac{\text{unsafe move}}{\text{correct move} + \text{unsafe move}}$, $\text{UAR} = \frac{\text{correct move}}{\text{total action}}$, $\text{MIR} = \frac{\text{missed intent}}{\text{total action}}$, and $\text{NRA} = \frac{\text{safe idle}}{\text{total rest}}$ for policies P0–P2 under each sensing scenario.

IV. RESULTS

A. Main LOSO performance under nominal sensing (S0)

Under nominal sensing (S0; all modalities available), TriSaFe-Trans achieves strong action decoding performance, with an accuracy of 0.903 ± 0.079 and balanced accuracy **0.882 ± 0.079** . The model attains a task accuracy of 0.727 ± 0.096 and balanced accuracy **0.680 ± 0.100** (Table I). Figure 6 shows the corresponding confusion matrices for the action and task decoding under S0 with P0. Compared to tri-modal baselines, TriSaFe-Trans achieves the best action balanced accuracy and competitive task balanced accuracy while supporting reliability-gated, safety-aware operation.

a) *Statistical significance vs. baselines:* Statistical significance was assessed using a paired Wilcoxon signed-rank test on the $n=20$ LOSO folds common to each comparison, with Holm correction across metrics (Table II). TriSaFe-Trans achieves positive median improvements on all metrics across all baselines, and significantly outperforms GRU and Hybrid on every reported metric ($p_{\text{Holm}} < 0.05$; RBC ≈ 0.58 –1.00). Against TCN, gains are Holm-significant for task decoding ($\max p_{\text{Holm}} = 4.25 \times 10^{-3}$) and action AUPRC ($p_{\text{Holm}} = 1.94 \times 10^{-2}$), while action Bal-Acc/Macro-F1/Acc improvements are positive but not Holm-significant.

TABLE I

MAIN LOSO PERFORMANCE UNDER SCENARIO S0 (ALL SENSORS AVAILABLE). VALUES ARE MEAN $\pm$ STD ACROSS SUBJECTS.

Model	Action head		Task head	
	Acc	Bal-Acc	Acc	Bal-Acc
TriSaFe-Trans (ours)	0.903 $\pm$ 0.079	0.882 $\pm$ 0.079	0.727 $\pm$ 0.096	0.680 $\pm$ 0.100
TriModal-GRU (baseline)	0.826 $\pm$ 0.127	0.825 $\pm$ 0.129	0.682 $\pm$ 0.145	0.558 $\pm$ 0.092
TriModal-TCN (baseline)	0.786 $\pm$ 0.162	0.784 $\pm$ 0.166	0.576 $\pm$ 0.206	0.478 $\pm$ 0.132
TriModal-Hybrid (baseline)	0.780 $\pm$ 0.118	0.779 $\pm$ 0.121	0.594 $\pm$ 0.150	0.410 $\pm$ 0.098

TABLE II

PAIRED WILCOXON SIGNED-RANK SIGNIFICANCE OF *TriSaFe-Trans* (OURS) VS. BASELINES ACROSS LOSO FOLDS (HOLM-CORRECTED).

Baseline	Folds	All $\Delta > 0$?	$\max p_{\text{Holm}}$	RBC	$\max p_{\text{Holm}}$ (task)	p_{Holm} (AUPRC)
GRU(all)	20	Yes	0.04276	0.581–0.857	0.04276	0.001829
Hybrid(all)	20	Yes	0.0007839	0.848–1.000	0.0001450	1.34×10^{-5}
TCN(all)	20	Yes	0.07993	0.543–1.000	0.004253	0.01944

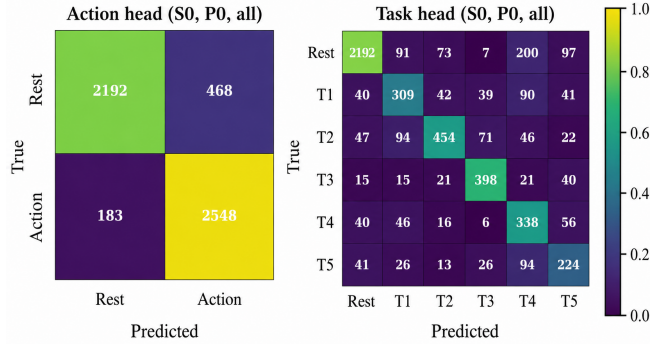


Fig. 6. Confusion matrices for action and task decoding (S0, P0). Left: binary action (REST vs. ACTION) predictions. Right: multi-class task predictions.

B. Modality ablations and multimodal synergy

Modality ablations under nominal sensing (S0) highlight the complementary roles of the three input modalities (Table III). EEG-only provides limited task specificity (task balanced accuracy 0.261 ± 0.050), EMG-only is strongest for detecting action (action balanced accuracy 0.827 ± 0.114), and ET-only is most informative for distinguishing tasks (task balanced accuracy 0.585 ± 0.128). Importantly, fusing all three modalities outperforms the best unimodal setting for both action and task decoding, indicating consistent multimodal synergy (Table III).

MSG is defined as the tri-modal balanced accuracy minus the best unimodal balanced accuracy. TriSaFe-Trans shows positive MSG for both action (0.055) and task (0.095), while some baselines yield negative action MSG (Table IV). These results show tri-modal synergy, and dropout robustness is summarized in Figure 7.

Robustness under single-modality failures is summarized in Table V. Relative to S0, TriSaFe-Trans exhibits the largest task degradation when EEG is removed (Δ task balanced

TABLE III

LOS0 BAL-ACC UNDER MODALITY ABLATIONS IN S0 (MEAN $\pm$ STD).

Model	Ablation	Action bal-acc	Task bal-acc
TriSaFe-Trans (ours)	eeg	0.685 $\pm$ 0.130	0.261 $\pm$ 0.050
	emg	0.827 $\pm$ 0.114	0.412 $\pm$ 0.075
	et	0.778 $\pm$ 0.172	0.585 $\pm$ 0.128
TriModal-GRU (baseline)	eeg	0.500 $\pm$ 0.000	0.167 $\pm$ 0.004
	emg	0.785 $\pm$ 0.154	0.354 $\pm$ 0.068
	et	0.760 $\pm$ 0.125	0.479 $\pm$ 0.114
TriModal-TCN (baseline)	eeg	0.500 $\pm$ 0.000	0.167 $\pm$ 0.004
	emg	0.827 $\pm$ 0.122	0.331 $\pm$ 0.077
	et	0.761 $\pm$ 0.127	0.444 $\pm$ 0.105
TriModal-Hybrid (baseline)	eeg	0.499 $\pm$ 0.002	0.168 $\pm$ 0.013
	emg	0.739 $\pm$ 0.101	0.280 $\pm$ 0.042
	et	0.735 $\pm$ 0.111	0.388 $\pm$ 0.087

TABLE IV

MULTIMODAL SYNERGY GAIN (MSG) OF TRI-MODAL FUSION OVER THE BEST UNIMODAL ABLATION.

Model family	MSG (action bal-acc $\uparrow$)	MSG (task bal-acc $\uparrow$)
TriSaFe-Trans (ours)	0.055	0.095
TriModal-GRU (baseline)	0.040	0.078
TriModal-TCN (baseline)	-0.043	0.034
TriModal-Hybrid (baseline)	0.040	0.022

accuracy = 0.302), with additional drops under EMG and ET dropout (0.146 and 0.236, respectively; Table V). For action decoding, EEG dropout has the largest impact (0.193), followed by EMG dropout (0.112), while ET dropout is negligible (0.002; Table V). These drops reflect the model's strong reliance on modality-specific cues under nominal sensing. Global modality usage entropy (GMUE) is reported under S0 as evidence that the model does not collapse onto a single modality (Table V); this gate-based reliability signal can be exploited by the policy layer to mitigate unsafe behavior when sensing degrades. The safety trade-off between unsafe executed-action fraction (SAE) and missed intent rate (MIR) across thresholds and dropout scenarios is

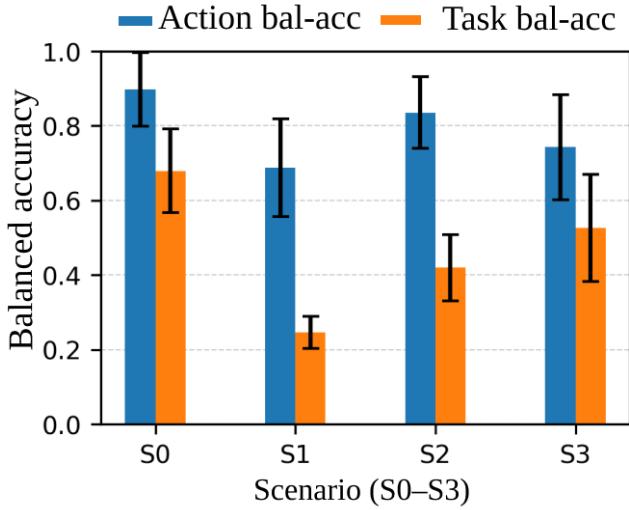


Fig. 7. Balanced accuracy of TriSaFe-Trans under nominal sensing and single-modality dropout scenarios.

TABLE V

ROBUSTNESS TO SINGLE-MODALITY DROPOUT (P0): Δ BAL-ACC FROM S0 TO S1–S3.

Dropped modality	Δ Action bal-acc↓				Δ Task bal-acc↓			
	Ours	GRU	TCN	Hybrid	Ours	GRU	TCN	Hybrid
EEG	0.193	-0.001	0.000	-0.007	0.302	0.000	0.000	-0.002
EMG	0.112	0.058	0.150	0.132	0.146	0.045	0.057	0.039
ET	0.002	0.061	0.073	0.089	0.236	0.252	0.232	0.155

GMUE (S0, all modalities, TriSaFe-Trans): 1.0364 ± 0.0070 .

shown in **Figure 8**.

C. Safety-performance trade-offs across policies

Policy-level safety evaluation in S0 is summarized in **Table VII**. In S0, P1 reduces to P0 because all modalities are available and $r_i = \sum_m s_m \gamma_{i,m} = \sum_m \gamma_{i,m} = 1$; differences emerge under S1–S3 when sensor-status gating decreases the weight of missing modalities. P1 is included for comparison, as it diverges from P0 under sensor-failure scenarios (S1–S3): modality availability alters r_i and therefore changes the effective execution score. Under P0 and P1, TriSaFe-Trans maintains a low missed intent rate (MIR = 0.055) but exhibits a relatively high unsafe executed-action fraction (SAE = 0.415), while the correct execution rate on ACTION windows is UAR = 0.637. The safety-first policy P2 reduces unsafe executed actions (SAE 0.415 $\rightarrow$ 0.155) and increases rest safety (NRA 0.819 $\rightarrow$ 0.975), at the cost of a lower correct execution rate on ACTION windows (UAR 0.637 $\rightarrow$ 0.313) and a higher missed intent rate (MIR 0.055 $\rightarrow$ 0.569; Table VII). Thus, P2 serves as a conservative safety upper bound. Increasing the execute/stop threshold τ makes the policy more conservative: the correct execution rate on ACTION windows (UAR) decreases, the missed intent rate (MIR) increases, and the unsafe executed-action fraction (SAE) decreases, yielding a consistent safety-responsiveness trade-off (see **Table VI**). This enables application-dependent

TABLE VI

EXAMPLE OPERATING-POINT SELECTION FOR P2 IN S0: SMALLEST τ SUCH THAT $\text{SAE}(\tau) \leq \delta$.

Safety budget δ	τ^*	SAE at τ^*	MIR at τ^*
0.15	0.30	0.093	0.694
0.10	0.30	0.093	0.694
0.05	0.35	0.035	0.793
0.03	0.40	0.005	0.875

TABLE VII

SAFETY METRICS FOR TriSaFe-TRANS IN S0 UNDER POLICIES P0–P2. VALUES ARE LOSO MEAN ACROSS FOLDS.

Policy	SAE↓	UAR↑	MIR↓	NRA↑
P0	0.415	0.637	0.055	0.819
P1	0.415	0.637	0.055	0.819
P2	0.155	0.313	0.569	0.975

operating points, allowing τ to balance fewer unintended activations against an increased risk of missed assistance.

Deployment involves selecting τ to meet a predefined safety budget. Specifically, the operating point is chosen as $\tau^* = \arg \min_{\tau} \text{MIR}(\tau)$ s.t. $\text{SAE}(\tau) \leq \delta$, where δ is the unsafe-execution budget. Table VI reports τ^* obtained from a threshold sweep on the validation split. TriSaFe-Trans has 1.201M parameters and runs in ≈ 10.9 ms per 500-sample window on a CPU (≈ 92 Hz), with essentially identical latency under S0 and S3.

D. Discussion and Limitations

Taken together, the results support three main observations. First, TriSaFe-Trans improves nominal LOSO performance over the compared baselines. Second, the ablation and robustness analyses show that EEG, EMG, and eye-tracking contribute complementary information, motivating the use of reliability-aware fusion rather than reliance on any single modality. Third, the policy analysis demonstrates a tunable safety-responsiveness trade-off, allowing the operating point to be adjusted according to deployment priorities. An important limitation is that the dataset was collected from healthy adults performing the tasks themselves. Accordingly, the present results should be interpreted as a non-clinical benchmark rather than direct validation in target assistive users, particularly because EMG may be weak or absent in real users. In practice, deployment for target users may more closely resemble a degraded-EMG setting, with greater reliance on EEG and eye-tracking.

V. CONCLUSION

TriSaFe-Trans is a safety-aware tri-modal transformer for EEG-, EMG-, and eye-tracking-based assistive intent recognition. Under nominal sensing, it improved LOSO action/task decoding over the compared baselines and showed positive multimodal synergy. Policy-level evaluation demonstrated an explicit safety-responsiveness trade-off across sensing conditions and execution policies. Future work will evaluate the pipeline with individuals with physical disabilities and

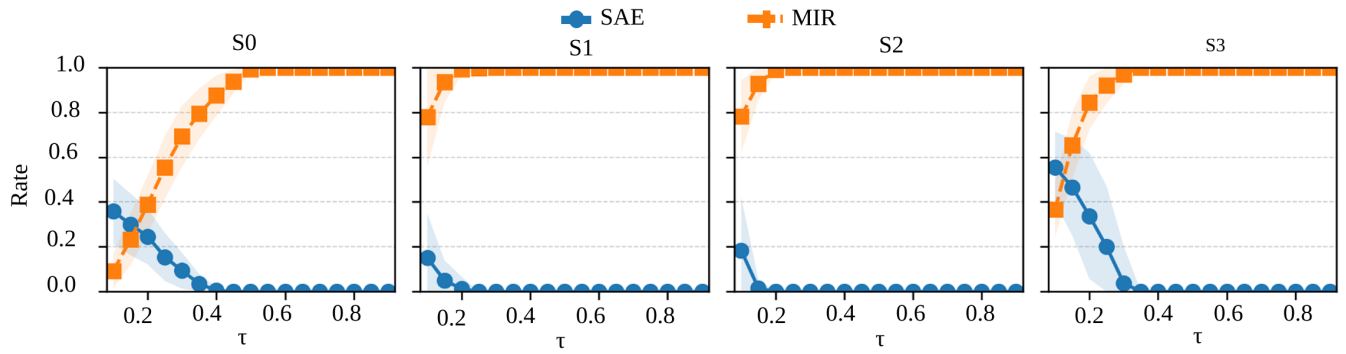


Fig. 8. Safety trade-off under single-modality dropout (S0–S3): unsafe executed-action fraction (SAE) and missed intent rate (MIR) versus the execute/stop threshold τ . S0: all sensors; S1: drop EEG; S2: drop EMG; S3: drop eye-tracking. Shaded bands show LOSO mean $\pm$ std.

incorporate transfer learning and subject-specific adaptation to better handle inter-user variability, weak or missing EMG, and real-world closed-loop deployment.

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